PART 1

Standard Operating Procedure (SOP)

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1. Purpose and scope

This SOP defines how authorised personnel start, stop, monitor and safely change the operating state of the Filler Cell PLC (PLC-FIL-02). It covers routine operation and online access with STEP 7. It does not cover machine-level mechanical procedures, which are in separate machine SOPs.

2. Roles and responsibilities

Role	Responsibility
Operator	Runs the line from the HMI. Does **not** use STEP 7 or change CPU mode.
Maintenance technician	Performs inspections, diagnostics, module and memory card handling under a work permit.
Controls engineer	Authorises program changes, downloads, firmware updates and password changes.
Shift supervisor	Approves LOTO release and restart after any PLC-related stop.

3. Safety prerequisites

- Only qualified personnel may work on the system, as required by the Siemens warning notices.
- Apply site lockout/tagout before touching terminals, removing connectors or installing/removing modules. Disconnect power before module work.
- The CPU has no physical RUN/STOP switch. Mode changes are made from STEP 7 (Start CPU / Stop CPU) or the CPU operator panel, so a STOP command can be issued remotely. Inform the operator before any mode change.
- Never force I/O values on a running line unless the controls engineer has approved the force in writing.

4. Normal start-up procedure

- Confirm the machine is in a safe state, guards closed and E-stop circuit healthy.
- Switch on the 24 VDC supply, then the CPU. The CPU performs startup (RUN/STOP LED flashes green/yellow alternately) and enters the configured start-up mode.
- Check the status LEDs: RUN/STOP solid green, ERROR off, MAINT off. LINK (green) and Rx/Tx (yellow) show PROFINET activity.
- Confirm HMI communication and that no diagnostic alarms are active.
- Release the machine for production only after the supervisor signs the start-up checklist.

5. Normal shutdown procedure

- Complete or purge the current batch and bring the line to its safe idle state from the HMI.
- If the program is to be stopped, move the CPU to STOP only after outputs are in a safe state. Review the configured behaviour of outputs on the RUN-to-STOP transition (substitute value or freeze last value) in device configuration.
- Isolate and lock out power if hardware work is planned.

6. Going online with STEP 7

- Connect the laptop to the CPU PROFINET port and verify the laptop IP is in the same subnet as the CPU.
- Use *Go online*, then compare offline and online projects before any download. If they differ, stop and call the controls engineer.
- Use watch tables for read-only monitoring by default. Modify values only with approval.
- Document every online session in the maintenance history (Part 4).

7. Program changes and downloads

- Back up the current project and a CPU upload before any change.
- Normal changes are downloaded in STOP mode during a planned stop. Download-in-RUN is allowed only for approved, low-risk block changes meeting the prerequisites in the manual, and requires supervisor consent.
- After download, verify the CPU returns to RUN, ERROR LED is off, and the diagnostic buffer has no new error entries.
- Update the project archive and record the change in Part 4.

8. Access protection

CPU access protection must remain enabled with a site-managed password stored in the approved vault. Shared or default passwords are not permitted. Password recovery requires a CPU factory reset, which erases the program, so a current project backup must always exist.

9. Records

Start-up checklists, online-session logs and download records are retained for 3 years in the maintenance record system.